MIL-S-5807C (ASG)
29 DECEMBER 1954
Superseding
MIL-S-5807B
12 August 1952

MILITARY SPECIFICATION

SEXTANT, AIRCRAFT, PERISCOPIC

This specification has been approved by the Department of the Air Force and by the Navy Bureau of Aeronautics.

1. SCOPE

1.1 This specification covers one type of periscopic sighting aircraft sextant, carrying case, and mount.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, drawings, and publications, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

QQ-A-277	Aluminum Alloy 75S Bars, Rods, and Shapes, Extruded
QQ-A-282	Aluminum Alloy 75S Bars, Rods, Shapes and Wire, Rolled or Drawn
QQ-P-416	Plating, Cadmium (Electrodeposited)

Military

MIL-A-8625	Anodic-Coatings, for Aluminum and Aluminum Alloys
MIL-C-5015	Connectors, Electrical, "AN" Type
MIL-D-3716	Dessicants (Activated) for Dynamic Dehumidification
MIL-D-5028	Drawings and Data Lists: Preparation of Manufacturers' (for Production Aircraft, Guided Missiles, Engines, Accessories, and Other Auxiliary Equipment)

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T05:47Z
Check the source to verify that this is the current version before use.

MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification for
MIL-F-5562	Fluid; Bubble Sextant
MIL-M-3171	Magnesium Alloy; Processes for Corrosion Protection of
MIL-P-6064	Packaging of Lightweight Aircraft Accessories
MIL-S-7742	Screw Threads, Standard, Aeronautical
MIL-W-530	Webbing, Textile, Cotton, General Purpose, Natural or in Colors
JAN-F-675	Films, Reflection-Reducing, for Glass Optical Elements

STANDARDS

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U. S. Military Property
MS28011	Sextant - Aircraft, Periscopic

DRAWINGS

Air Force-Navy Aeronautical Standard Drawings

AN5740	Watches; Navigation, Master
AN5742	Watch - Navigational Stop
AND10398	Metals - Definition of Dissimilar

PUBLICATIONS

Air Force-Navy Aeronautical Bulletins

No. 143	Specifications and Standards; Use of
No. 405	Storage Life - Aeronautical Articles

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Component parts.- This specification covers a sextant and a mount for supporting the sextant and providing a pressurized seal. The sextant, complete with carrying case and spares, shall consist of the following:

(a) 1 Sextant.
(b) 1 carrying case.
(c) 1 cable, electrical connecting.
(d) 2 spare lamps.
(e) 1 watch clip.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T05:47Z
Check the source to verify that this is the current version before use.

3.2 Materials.-

3.2.1 Metals.- Metals shall be of the corrosion-resistant type or shall be suitably protected to resist corrosion during normal service life.

3.2.2 Dissimilar metals.- Dissimilar metals, as defined by Drawing AND10398, shall not be used in intimate contact with each other.

3.2.3 Nonmagnetic materials.- Nonmagnetic materials shall be used for all parts of the averagers, except where magnetic materials are essential.

3.2.4 Fungus-proof materials.- Materials which are not nutrients for fungi shall be used to the greatest extent practicable. In cases where materials that are fungus nutrients must be used, such materials shall be either treated with a fungicidal agent as approved by the procuring activity or be sealed in a fungus-proof chamber. Fungus nutrient materials used on the outside of the sextant must be suitably treated unless such treatment would be detrimental to the operation of the sextant.

3.2.5 Protective treatment.- When materials are used in the construction of the sextants that are subject to corrosion in salt air or other atmospheric conditions which may occur during service usage, they shall be protected against such corrosion in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of atmospheric conditions shall be avoided.

3.2.6 Selection of materials.- Specifications and standards for all materials, parts, and Government certification and approval of processes and equipment, which are not specifically designated herein and which are necessary for the execution of this specification, shall be selected in accordance with ANA Bulletin No. 143, except as provided in the following paragraph.

3.2.6.1 Standard parts.- Standard parts (MS, AN, or JAN) shall be used wherever they are suitable for the purpose, and shall be identified on the drawing by their part numbers. Commercial utility parts such as screws, bolts, nuts, cotter pins, etc, may be used, provided they possess suitable properties and are replaceable by the standard parts (MS, AN, or JAN) without alteration, and provided the corresponding standard part numbers are referenced in the parts list and, if practicable, on the contractor's drawings. In the event there is no suitable corresponding standard part in effect on date of invitation for bids, commercial parts may be used provided they conform to all requirements of this specification.

3.3 Design and construction.-

3.3.1 Sextant.- Hereinafter, the term "sextant" shall be construed to mean the sextant proper, periscopic tube, connecting cable, and watch clip. The sextant shall be a bubble-type sextant built in the form of a periscopic telescope with the periscope projecting above the skin of the aircraft. Provisions shall be made for 360 degrees rotation of the instrument around the vertical axis and tilt of the sextant up to 14-1/2 degrees minimum from the vertical axis. The sextant shall conform essentially to Standard MS28011.

3.3.1.1 Case.- The case shall consist of a body enclosing all parts of the sextant normally enclosed in instruments of this type.

3.3.1.2 Body.- The body shall be constructed of either nonferrous low-density metal or synthetic material, uniform in texture with a smooth surface. Synthetic material shall be of the thermalsetting type, composed of a suitable filler and phenol condensation product binder. The body shall be finished in durable dull black.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T05:47Z
Check the source to verify that this is the current version before use.

3.3.1.3 Bubble.-

3.3.1.3.1 Bubble cell replacement.- The sextant and mount shall be so designed and constructed that replacement of the bubble cell, together with the operation necessary to bring the sextant into complete operative adjustment, may be readily accomplished without disturbing alinement of other parts of the sextant. The bubble cell shall be accurately constructed in order to provide interchangeability.

3.3.1.3.2 Illumination of bubble image.- Artificial illumination giving a dark field shall be provided through a pilot light in the bubble chamber operating on approximately 28 volts ac or dc and so designed that the bulb is easily renewable. The illumination of the bubble shall be such that the bubble image is clearly defined with minimum glare in the field to permit adequate dimming of the illumination for observation of second magnitude stars. Provision shall be made for suitable illumination of the edge of the bubble field. Where the bubble is exposed to daylight, a diffuser shall be provided which shall be capable of being swiveled away from the line of sight to the scale.

3.3.1.3.3 Bubble adjustment.- The bubble shall be capable of being formed and adjusted in size when the sextant is in the "Up" position in the mount.

3.3.1.3.4 Bubble liquid.- The liquid used in the bubble level shall conform to Specification MIL-F-5562. Other liquids may be used if specifically approved by the procuring activity.

3.3.1.4 Optical requirements.- The optical requirements shall be as follows.

3.3.1.4.1 All lenses, prisms, and mirrors shall be of first quality, made from first-class stock and shall be accurately ground, polished, and centered. The optical system shall be well corrected for color and spherical aberration. Lenses shall be mounted free from strain.

3.3.1.4.2 The exit pupils for the bubble and for the star field shall be concentric within 1/2 millimeter when viewed from a point on the ocular axis extended and shall appear to be concentric with the eye lens.

3.3.1.4.3 The eye distance for the star field and for the bubble field shall not differ by more than 3 millimeters.

3.3.1.4.4 The eyepiece shall be adjustable for focus. The focusing arrangement shall be easily manipulated, and shall be sufficiently positive in order that slight jars experienced during observation in an airplane will not disturb the setting.

3.3.1.4.5 The index prism, shall not have pyramidal error in excess of 30 seconds.

3.3.1.4.6 The image of the celestial body, true heading scale, and the bubble shall be erect.

3.3.1.4.7 The image of the celestial body and the bubble shall be brought into coincidence by reflection.

3.3.1.4.8 Magnification of the optical system shall be two-power with adjustable focus from -2 to +2 diopters.

3.3.1.4.9 The sextant shall have a real field of not less than 14.5 degrees, definition not being critical beyond 5 degrees real field from the sighting axis. All internal optics except the beam splitter, etched reticle surface, and bubble chamber surfaces, shall be treated with a high-efficiency nonreflecting coating. The coating of exposed surfaces shall be abrasion resistant in accordance with Specification JAN-F-675.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T05:47Z
Check the source to verify that this is the current version before use.

3.3.1.4.10 The angle of sight (eyepiece) shall be horizontal, and shall lie in the same or parallel vertical plane as the line of sight of the exit viewing prism.

3.3.1.4.11 The exit pupil shall be not less than 4.5 millimeters.

3.3.1.4.12 Magnification of the optical system for viewing the true heading scale shall be at least two-power with adjustable focus. The design shall be such that no parallax exists between the scale and cross hair images. At least 25 degrees of the true heading scale shall be visible in the eyepiece at all times.

3.3.1.4.13 Foreign particles (bubble).- Any loose particles in the bubble cell or any evidence of stickiness or erratic behavior of the bubble during any test in this specification shall be considered cause for rejection.

3.3.1.4.14 Foreign particles (field of view).- Every effort shall be made to keep foreign particles in the field of view to a minimum. The maximum number of foreign particles permitted in the field of view shall be determined as follows:

> Scanning the field of view, any particle larger than 0.004 shall be cause for rejection. The remaining particles are to be counted and the count weighed as follows:
>
> Particles over 0.001, count as 10
> Particles 0.0003 to 0.001, count as 5
> Particles under 0.0003, count as 2.5

Particle size may be estimated by comparison to the etched reticle lines which are 0.0008 wide.

In the center zone of the field of view, defined as a circle of 5 degrees true field, the total weighed count of the particles shall not exceed 5. The remainder of the field of view, the outer zone, shall not have particles with a weighed count exceeding 30. The center zone may be approximated by comparison to the entire field of view which is 15 degrees true field.

3.3.1.5 Cross hairs.- The optical system of the sextant shall contain horizontal and vertical cross hairs centered in the line of sight to facilitate collimation and the reading of true heading. The cross hairs, bubble image, star image, and scale image shall all have a common focal plane. The cross hairs shall be visible and distinct when subjected to the artificial lighting of the bubble cell.

3.3.1.6 Altitude range.- The sextant shall be capable of measuring altitudes from 10 degrees below the horizontal (zero) to 2 degrees above 90 degrees altitude. The altitude shall be indicated on a drum-type counter with one complete revolution of the operating knob indicating approximately 5 degrees change in altitude.

3.3.1.6.1 Altitude counter.- An altitude counter mechanism shall be incorporated. The counter shall be so located that it can be read from the right side of the sextant. The counter and averaging windows shall be located close together and so illuminated that they may be read simultaneously.

3.3.1.6.2 Counter range.- The counter shall indicate altitudes throughout the effective range of not less than -10 degrees to +92 degrees. Rugged stops shall be provided to prevent damage to the sextant, or to prevent disengagement of the operating parts when the operating knob is rotated in either direction beyond the operating range of the sextant. The design of the stops shall be such that forcible engagement of the stops imposes no pressure between applicable operating parts of the sextant mechanism other than the normal pressure existing when the stops are disengaged.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T05:47Z
Check the source to verify that this is the current version before use.

3.3.1.6.3 Degree indicators.- The counter shall consist of three adjacent drums. The first two drums shall contain numerals from 0 through 9, with a height of approximately 5/32 inch. The numerals and markings on the drums shall be of durable dull black. The numerals on the first two drums shall indicate altitude in degrees.

3.3.1.6.4 Minute indicator.- The third drum shall contain numerals for each 5 minutes of altitude from 0 through 55. The height of these numerals shall be 3/32 inch. Graduation marks for each minute of altitude shall also appear on the third drum. The graduation marks for each minute of altitude shall be approximately 3/64 inch in length, with each consecutive 5-minute graduation mark approximately 5/64 inch in length, and both marks shall not exceed 0.005 inch in width.

3.3.1.6.5 Index line.- An index line engraved in durable dull white of 1/4-inch length and 0.01-inch width shall be vertically centered on the counter cover glass over the minute indicator, the cover glass in turn centered with respect to the counter drums. The index line shall be so situated to afford rapid alinement of the degree and minute values for accurate altitude reading.

3.3.1.7 Filters and shade glasses.- Suitable filters shall be provided and shall have a visible light transmission through a range of approximately 0.001 percent to 85.0 percent. Introduction of the shade glass into the line of sight shall not produce secondary images or deviation in excess of 3 minutes of arc in horizontal plane or 30 seconds of altitude in a vertical plane.

3.3.1.8 Sextant illumination.- The sextant shall contain adequate provisions to illuminate the watch, altitude counters, bubble, and true heading scale. The lamps, furnished in accordance with Standard MS28011, shall be connected in parallel, and shall be controlled by a toggle switch on the mount. The bubble chamber lamp shall also be controlled by a rheostat which does not contain an off-on switch. The illumination from these lamps shall cause no glare in the field of view. The rheostat shall be capable of reducing bubble illumination to the point which permits taking observation on second magnitude stars. A receptacle as shown on Standard MS28011 shall be installed on the sextant to receive power from the mount via the connecting cable. The receptacle shall be constructed of 75S-T6 aluminum in accordance with Specification QQ-A-277 or QQ-A-282. The true heading scale shall be illuminated to permit adequate visibility when taking observations on second magnitude stars. The bubble and true heading scale shall be adequately illuminated for any combination of outside bright daylight and inside dim aircraft illumination levels.

3.3.1.9 Connecting cable.- The sextant shall be provided with a suitable connecting cable with connector plugs on both ends in accordance with Standard MS28011.

3.3.1.10 Watch clips.- A watch clip permanently fastened to the sextant, suitable for securely holding the AN5740 or AN5742 watch, shall be so positioned on the sextant proper as to allow the watch face to be read with the eye in line with the eyepiece position. The watch shall be illuminated to permit adequate readability.

3.3.1.11 Variable time averager (VTA).- The variable time averager shall be designed to continuously average sextant altitude settings, and shall give the average altitude and half the elapsed time when used over any time interval from 30 seconds to 2 minutes. The averager shall be so constructed that no parts will work loose in service and shall be so built to withstand the strains, jars, vibrations, and other conditions incident to shipping, storage, installation, and service, without failure. Backlash shall be reduced to a minimum.

3.3.1.11.1 Dust protection.- The averager shall be enclosed in a dustproof housing.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T05:47Z
Check the source to verify that this is the current version before use.

3.3.1.11.2 Winding lever.- The winding lever shall be located conveniently for winding the mechanism. One complete turn or less shall be sufficient to set the altitude averaging indicators to their respective zero positions. Where a shutter is used in the system, zero setting of the winding lever shall remove the shutter from the line of vision. Approximately 5 seconds after release of the winding lever, the averager shall be ready for operation. The method of returning indicator to zero position shall not involve forced slippage of any part of the averager.

3.3.1.11.3 Averager zero adjustment.- After the mechanism has been rewound and the averager made ready for its intended usage, it shall be possible to adjust accurately the zero position of the averager by means of an adjustment screw accessible either from the outside of the case, or by removing the cover of the averager without disassembly of the mechanism.

3.3.1.11.4 Half-time dial.- The diameter of the half-time dial shall be a minimum of 3/4 of an inch. It shall be graduated into 60 equal divisions, each division to represent 2 seconds of time. Every fifth graduation shall be a major graduation, identified by numerals. The numerals shall read from 0 to 55. The dial shall indicate elapsed time to within an accuracy of ±4 seconds, or ±2 seconds if read on the half-time dial. The timing mechanism shall drive the half-time dial to indicate elapsed time over a period of 2 minutes with a maximum allowable error of 2 seconds per minute.

3.3.1.11.5 Averaging indicator.- An indicator shall be provided which, when returned to its zero position by operation of the sextant control knob upon completion of an observation and prior to rewinding, shall determine the average altitude, as read on the sextant altitude counter, to within 2 minutes of arc after a 2-minute observation without ambiguity within 40 degrees throughout the total sextant altitude range.

3.3.1.11.6 Dial marking.- The markings of the half-time dial and the altitude averaging indicators shall have a background of durable dull black finish. Numerals and reference markings shall be painted white. When a revolving disk-type indication is used, a white rectangle approximately 1/8 by 1/4 of an inch shall be used in addition to the durable dull black for the background of the reference markers of the averaging indicator. The index mark on the white rectangle shall be a dull black.

3.3.1.11.7 Operating button or lever.- A single start-stop button or lever shall start and stop the operation of the averager. The button shall be so located that when an observer holds and sights through the sextant having the averager attached, he may conveniently operate the button with the thumb or one of the first two fingers of the hand not being used to operate the altitude control knob of the sextant.

3.3.1.11.8 Period of operation.- When the averager is fully wound and ready for operation, the mechanism shall start and continue functioning when the start-stop button is moved through not more than 3/4 full displacement. The button may then be released without affecting the operation of the mechanism. The averager shall continue in operation through a period of 2 minutes ±4 seconds and after operating for the maximum period, the averager shall stop automatically. After the averager has been operating for a period not to exceed 30 seconds, depressing and releasing the button at any time before the total period has elapsed shall stop the operation.

3.3.1.11.8.1 A shutter shall automatically intercept the light from the observed body when the start-stop button is depressed to stop operation, or when the cycle of operation of the averaging device is complete. Within the period of 2 seconds of the end of the averager cycle, the averager shall actuate the shutter mechanism.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T05:47Z
Check the source to verify that this is the current version before use.

3.3.1.11.9 Range.- The averager shall be capable of averaging a series of observations extending over the full sextant altitude range without limit stops.

3.3.1.11.10 Windows.- Suitable plastic windows shall cover both the half-time dial and the averaging indicators. An index line 0.01 inch wide shall be vertically centered on the window with respect to the zero position of the averaging indicators and half-time dial. That portion of the line extending over the averaging disks shall be durable dull black, while that portion of the line extending over the half-time dial shall be durable dull white.

3.3.1.11.11 Eye guard.- The eyepiece shall be provided with a suitable rubber eye guard to protect the eye and to exclude light. The eye guard shall be so designed that it may be readily adjusted without tools for use with either eye and shall be free to rotate under service conditions. A positive means of attaching the rubber eye guard shall be provided. Cementing alone shall not be considered adequate.

3.3.1.11.12 Averager instruction plate.- An instruction plate carrying brief instructions for the operation of the averager shall be placed on its outside housing. The instructions on the plate shall be subject to the approval of the procuring activity.

3.3.1.12 Moisture indicator.- Visible silica gel indicators shall be provided in the sealed tube and case to indicate the presence of moisture. The silica gel provided shall be in accordance with Specification MIL-D-3716, type IV, group H.

3.3.2 Mount.- The sextant mount shall conform to Standard MS28011. The sextant mount shall be designed and constructed to receive the periscopic tube of the sextant and shall allow the sextant to be positioned "Up" for sighting, and "Down" when the sextant is not in use. The sextant shall be capable of being removed from the mount. Provisions for tilting the sextant proper at least 14-1/2 degrees from the vertical in any direction shall be contained in the mount. When mounted on a flat surface, no part of the mount shall touch the mounting surface at full tilt.

3.3.2.1 Pressure seal.- The mount shall be designed to provide a mechanism to seal the opening in the aircraft skin or astro sighting window with a flush seal when the sextant is in the "Down" position. A means shall be provided in the form of a safety lock or a series of positive releasing devices to require interrupted motions in removing the sextant from the mount. One interruption in the movement shall occur when when the sextant is lowered from the "Up" or operating position to the "Down" or retracted position. Another interruption in the movement shall occur when removing the sextant from the mount. When mounted in a suitable test chamber, the mount and cover shall be capable of withstanding a pressure differential of 20 psi without structural failure and shall restrict the loss of air in either the "Up" or "Down" position of the sextant to a maximum of 0.20 pound of air by weight per minute when the air in the test chamber is maintained at a constant differential pressure of 10 psi and an approximate air temperature of 100°F. The requirements of 0.20 pound of air by weight per minute is a maximum, and every effort should be taken to keep the leakage below this figure.

3.3.2.2 Azimuth scale.- Incorporated as an integral part of the mount shall be an azimuth scale reading through 360 degrees, and one lubber line. The azimuth scale shall be rotatable throughout 360 degrees, and the rotation shall be recorded by a digital counter in degrees and tenths of a degree. The first three numerals on the counter shall be durable dull white with a background of durable dull black. The tenths digit shall be of reversed or distinctive coloring. An optical system shall be provided by means of which an image of the azimuth scale be projected on the focal plane of the eyepiece in coincidence with the reticle. The relationship between the counter and the graduations of the scale shall be so adjustable that when the true azimuth of a celestial body is set on the counter, the true heading of the aircraft shall coincide with the reticle when sighting on the celestial body. Construction of the instrument shall be such that the over-all error of this instrument, when used for determination of true heading shall not exceed 1/4 degree . The azimuth scale shall be graduated in 1/2 degrees with numerals every 10 degrees. The scale and counter shall be illuminated for night operation. Provisions shall be made for rendering the scale invisible by

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T05:47Z
Check the source to verify that this is the current version before use.

blocking the illumination without turning off the bubble-scale lamp or by blocking the optical system. Rotation of the azimuth scale as recorded in the counter shall be accomplished by means of a crank assembly. A clockwise rotation of the crank shall increase the azimuth setting in the counter. The torque required to rotate the azimuth crank shall not exceed 10 in.-oz at any point of the travel of the azimuth scale.

3.3.2.3 Mount illumination.- The mount shall contain two receptacles as shown on Standard MS28011. Both receptacles shall be mounted on a nonrotating part of the mount. One of these receptacles shall receive power from the aircraft power supply to supply current to the counter illumination lamp in the mount and also to the second receptacle. The latter receptacle shall be so located and installed as to point downward when the mount is installed. This latter receptacle shall be constructed of 75S-T6 aluminum in accordance with Specification QQ-A-277 or QQ-A-282. The counter illumination lamp and receptacle furnishing power to the sextant shall be controlled by a toggle switch. Both lamp and toggle switch shall be as shown on Standard MS28011. The second receptacle shall be capable of furnishing power to the sextant via the connecting cable. The illumination from the lamp shall cause no glare in the field of view. The azimuth scale shall be adequately illuminated for any combination of outside bright daylight and inside dim aircraft illumination levels.

3.3.2.4 Mount pins.- The pins holding the ball joints or gimbals in place on the mount shall be securely anchored.

3.3.2.5 Sextant holding pins.- The pins used for holding the sextant in the mount shall be of such size and material as to preclude breakage while the sextant is held indefinitely in the mount in the "Up" or "Down" position during any normal flight conditions.

3.3.2.6 Drain plug.- A drain plug shall be provided at the lowest point of the shutter assembly. It shall be easily removable without tools or coins and shall be attached to the mount with a suitable retaining chain, or other mechanical means, to prevent loss.

3.3.2.7 Mounting screws.- Sufficient mounting screws shall be provided for attaching the sextant mount to the airplane fuselage. A minimum of two spare screws shall be provided.

3.3.2.7.1 An envelope containing the mounting screws shall be furnished with each sextant mount. The following information shall be printed on the face of the envelope.

IMPORTANT

THIS ENVELOPE CONTAINS MOUNTING
SCREWS FOR THE PERISCOPIC SEXTANT
MOUNT

3.3.3 Carrying case.- The carrying case shall be of rigid or semirigid construction using the lightest weight waterproof nonmagnetic materials consistent with the strength requirements listed below and shall be provided with a carrying handle, lid fastener (or slide fastener if desired when a semirigid case is provided). The dimensions of the case shall be in accordance with Standard MS28011. (Note: Since all dimensions are maximum, the entire case may be made smaller or portions of the case may be made smaller than indicated, in order that the shape of the case might conform to the shape of the sextant.) The weight of the case shall not exceed 8-1/2 pounds, and every effort shall be made to keep the weight below this figure.

3.3.3.1 Provisions shall be made within the carrying case to accommodate, in a snug manner, the sextant, cable, and spare lamps. This equipment shall be secured within the case.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T05:47Z
Check the source to verify that this is the current version before use.

3.3.3.2 The internal supports shall be constructed of sponge or foam rubber and shall be arranged within the case to furnish maximum support for the sextant. With all components in place within the closed carrying case, no movement shall be detected when the case is shaken vigorously. The area of contact between the supporting rubber and the sextant and components shall be sufficient to prevent damage to the contents at any test temperature when the case is handled roughly, such as being dropped on a firm solid surface from a height of 30 inches in any attitude or position. Guards may be added to the corners or edges of the case. Over-all dimensions shall not exceed those shown on Standard MS28011.

3.3.3.3 A card shall be attached within a holder to the inside of each case containing spaces for inserting sextant serial number, sextant index error at three points 0, 45, and 90 degrees, and the date of collimation of the sextant. Prior to delivery, this information shall be inserted on the card for each sextant.

3.3.3.4 Loops shall be provided on the case as shown on Standard MS28011 to position and guide tie-down webbing used to secure the case. The loops shall be capable of holding 1-inch webbing, type III, procured in accordance with Specification MIL-W-530.

3.3.3.5 When the sextant case is stowed in an upright position, the lid or fastener shall open without interference with the shelf or deck on which it is stowed. This may be accomplished as shown on Standard MS28011.

3.3.3.6 The carrying case shall be suitably marked to indicate the proper position for ground storage and for aircraft stowage.

3.3.4 Basic alinement.- The telescope must be within 2 degrees of the true vertical when the bubble and the cross hairs are in perfect alinement.

3.4 Interchangeability.- All parts having the same manufacturer's part number shall be directly and completely interchangeable with each other with respect to installation and performance. Changes in manufacturer's part numbers shall be governed by the drawing number requirements of Specification MIL-D-5028.

3.5 Finish.-

3.5.1 Aluminum-alloy parts.- Aluminum-alloy parts shall be covered with an anodic film conforming to Specification MIL-A-8625, except that small holes need not be anodized.

3.5.2 Steel parts.- Steel parts shall be cadmium plated where practicable, in accordance with Specification QQ-P-416, type II or III, as applicable, and of a class that is adequate to achieve the degree of protection required.

3.5.3 Magnesium-alloy parts.- All magnesium or magnesium-alloy parts shall be treated for corrosion resistance in accordance with applicable method of Specification MIL-M-3171.

3.6 Screw threads.- Unless otherwise specified, the threads of all machine screws, 0.060 inch or larger in diameter, shall conform to Specification MIL-S-7742.

3.7 Weight.- The weight of the sextant completely assembled shall not exceed 7-1/2 pounds. The weight of the mount shall not exceed 4-3/4 pounds.

3.8 Electrical connectors.- The electrical connectors shall be in accordance with Specification MIL-C-5015, except where otherwise noted.

3.9 Performance.- The sextant, mount, and carrying case shall meet the performance requirements specified in Section 4 when subjected to the applicable tests.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T05:47Z
Check the source to verify that this is the current version before use.

3.9.1 Shutter performance.- The sextant and projection lens shutters shall perform satisfactorily during all applicable tests of this specification.

3.10 Markings.- All scale graduations, numerals, and index marks shall be permanently and legibly marked. Engraved or etched markings shall be filled with enamel of color that will permit distinct reading. All identification and markings shall be engraved or etched and enamel-filled as specified.

3.11 Identification of product.- Equipment, assemblies, and parts shall be marked for identification in accordance with Standard MIL-STD-130, except that the data shall be as follows.

3.11.1 Sextant nameplate.-

SEXTANT, AIRCRAFT, PERISCOPIC
MS28011 (Appropriate dash number)
28 Volts ac or dc
Specification MIL-S-5807C
Manufacturer's Part No.
Manufacturer's Serial No.
Contract or Order No.
Stock No.
Manufacturer's name or trade-mark
US

3.11.2 Mount nameplate.-

MOUNT, PERISCOPIC SEXTANT
MS28011 (Appropriate dash number)
28 Volts ac or dc
Specification MIL-S-5807C
Manufacturer's Part No.
Manufacturer's Serial No.
Contract or Order No.
Stock No.
Manufacturer's name or trade-mark
US

3.11.3 Carrying case nameplate.-

SEXTANT, AIRCRAFT, PERISCOPIC
MS28011 (Appropriate dash number)
28 Volts ac or dc
Specification MIL-S-5807C
Manufacturer's Part No.
Contract or Order No.
Stock No.
Manufacturer's name or trade-mark
US

3.12 Service instructions.- Unless otherwise specified, one copy of commercial operation and installation instructions shall be included with each sextant and mount. Prior to publication, these instructions shall be approved by the procuring activity.

3.13 Workmanship.- The sextant, including all parts and accessories, shall be constructed and finished in a thoroughly workmanlike manner. Particular attention shall be given to neatness and thoroughness of soldering, wiring, marking of parts and assemblies, and freedom from burrs and sharp edges.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T05:47Z
Check the source to verify that this is the current version before use.

3.13.1 Dimensions.- Dimensions and tolerances not specified shall be as close as is consistent with the best shop practices. Where dimensions and tolerances may affect the interchangeability, operation, or performance of the sextant, they shall be held or limited accordingly.

3.13.2 Screw assemblies.- Assembly screws and bolts shall be tight. The word "tight" means that the screw or bolt cannot be appreciably tightened further, without damage or injury to the screw or bolt or threads. All screws and bolts shall be equipped with an appropriate locking medium.

3.13.3 Gears.- Gear assemblies shall be properly alined and meshed, and shall be operable without interference, tight spots, loose spots, or other irregularities. Where required for accurate adjustments, gear assemblies shall be free from backlash.

3.13.4 Cleaning.- The sextant shall be thoroughly cleaned of loose, spattered, or excess solder, metal chips, and other foreign material, after final assembly. Burrs and sharp edges, as well as resin flash that may crumble, shall be removed.

3.13.5 Pivots, bearings, and gears shall neither bind nor shake and shall be as near frictionless as practicable.

4. QUALITY ASSURANCE PROVISIONS

4.1 Classification of tests.- The inspection and testing of the sextants and mounts shall be classified as follows:

(a) Preproduction tests: Preproduction tests are those tests performed on samples representative of the production of the item after the award of contract, to determine that the production meets the requirements of this specification.

(b) Acceptance tests: Acceptance tests are those tests performed on individual lots which have been submitted for acceptance.

4.2 Test conditions.-

4.2.1 Standard atmospheric conditions.- Whenever the pressure and temperature existing at the time of test are not specified definitely, it is understood that test is to be made at atmospheric pressure (approximately 29.92 inches Hg) and at room temperature (approximately 25°C). When tests are made with atmospheric pressure or room temperature differing materially from the above values, proper allowances shall be made for the difference from the specified condition.

4.2.2 Attitude.- Unless otherwise specified, the sextant shall be tested in normal operating position.

4.2.3 Rain.- For purposes of simulating actual service conditions, the sextant shall be subjected to the Rain test as follows: With the sextant mounted in the mounting bracket and in the "Up" position, only the parts protruding through the skin of the aircraft, i.e., exposed to the atmosphere outside the fuselage of the aircraft, shall be subjected to tests. In addition, the mount with its cover closed shall be subjected to the Rain test.

4.2.4 Rate of altitude control knob rotation.- Unless otherwise specified, the rate of rotation of the drive mechanism for operating the sextant knob shall in no case exceed 10 degrees of altitude reading per second.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T05:47Z
Check the source to verify that this is the current version before use.

4.2.4.1 The term "knob," when used, has reference to the sextant altitude control knob or the equivalent rotating altitude control device on the test fixture. The term "altitude setting," when used, is assumed to mean the provision made on either the sextant or test fixture for reading the angular altitude set in by the altitude control knob.

4.2.5 Test measurements.- All measurements shall be made with instruments whose accuracy has been verified. If tests are conducted at the contractor's plant the accuracy of the instruments and test equipment shall be verified periodically by the contractor to the satisfaction of the procuring activity.

4.2.5.1 Tolerances.- The maximum allowable tolerances on test condition measurements shall be as follows:

(a) Temperature: Plus or minus 2°C or 3°F, as applicable.
(b) Altitude: Plus or minus 5 percent in feet.
(c) Humidity: Plus or minus 5 percent relative.
(d) Vibration amplitude: Plus or minus 5 percent. This tolerance is applicable only to the amplitude measuring instruments.
(e) Vibration frequency: Plus or minus 2 percent. This tolerance is applicable only to the frequency measuring instruments.
(f) Additional tolerances: Additional tolerances shall be as specified.

4.2.6 Test equipment.- When a special chamber is required to accomplish the tests, the equipment shall be mounted in the chamber and electrical or mechanical connections completed where operation is required during the test period. Access covers and inspection plates shall be in place except when such covers would normally be removed for service use. When operation is not required, external connections may be plugged to simulate actual service conditions.

4.2.7 Pressure.- A pressure chamber shall be used which is capable of withstanding a pressure differential of 20 psi continuously without loss of pressure. Provisions shall be made for mounting the periscopic sextant mount and seal in the chamber, with the sextant in the normal operating position or removed. A suitable air flowmeter shall be used to measure the flow of air to the test chamber.

4.2.8 Averager.- Unless otherwise specified, the averager shall be tested while attached to the sextant.

4.3 Preproduction tests.-

4.3.1 Sampling instructions.- The Preproduction test samples shall consist of two sextants with cases and mounts of each manufacturer's part number upon which testing is desired. Samples shall be identified as required and forwarded to the activity designated in the contract. Enclosed with the Preproduction test samples shall be a copy of the certified test report indicating that the items covered by this specification have been subjected to and have passed all the tests of this specification.

4.3.2 Tests.- The Preproduction tests of sextants and mounts shall consist of all the tests described under Individual tests and Sampling tests, and the following described tests which shall be accomplished after completion of the Sampling tests. Upon completion of each Preproduction test described below, the sextants and mounts shall be subjected again to the Individual and Sampling tests specified herein.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T05:47Z
Check the source to verify that this is the current version before use.

4.3.2.1 Vibration failure.- The test shall be conducted as described herein: First, with the sextant installed in a mount in the "Up" position and then with the sextant in the carrying case. The mount shall be vibrated by itself, and then with the sextant installed. Vibratory motions made on the sextant while installed in the mount shall be transmitted by the mount to the sextant when the mount is vibrated. Upon completion of the test, the sextant shall be subjected to and satisfactorily pass the Individual and Sampling tests. No looseness or damage to the mechanism shall result from the vibration. Resonant frequencies shall be determined by varying the frequency of applied vibration slowly through the frequency range of 10 to 500 cps at any vibratory accelerations not exceeding those shown in figure 1. This procedure shall be followed successively for vibration applied along three mutually perpendicular axes of the test specimen. The test specimens shall be vibrated at the indicated resonant conditions for 60 minutes with the applied amplitude or vibratory acceleration specified in figure 1. These periods of vibration shall be accomplished in sequence for each of three mutually perpendicular axes of vibration. When more than one resonant frequency is encountered with vibration applied along any one axis, the test period may be accomplished at the most severe resonance, or the period may be divided among the resonant frequencies, whichever is considered most likely to produce failure. When resonant frequencies are not apparent within the specified frequency range, the specimens shall be vibrated for a period of 2 hours at a frequency of 55 cps and an applied amplitude of 0.036 inch.

4.3.2.2 High temperature exposure.- The sextant shall be subjected to a temperature of 160°F (71°C) for a period of 24 hours. At the end of the 24-hour period and while still at the high temperature, the sextant shall operate without indications of sticking or erratic action. After the chamber has been returned to room temperature and the sextant has saturated at room temperature for a minimum period of 4 hours, it shall meet the Individual and Sampling test requirements specified herein. There shall be no damage which would adversely affect subsequent operation. At the discretion of the testing activity, the Individual tests may be performed immediately after removing the sextant and mount from the high temperature chamber.

4.3.2.3 Low temperature exposure.- The equipment shall be placed within the chamber and cooled to and maintained at a temperature of -85°F (-65°C) for a period of 48 hours. The temperature shall then be changed to -65°F (-54°C) and maintained for an additional 24-hour period or until stabilization is reached, whichever is longer. While at this temperature, the sextant shall be operated throughout its range without exhibiting sticking or erratic characteristics. The sextant shall then be subjected to the Individual and Sampling tests specified herein while at the -54°C temperature. There shall be no damage which would adversely affect subsequent operation. At the discretion of the testing activity, Individual tests may be performed immediately after removing the sextant and mount from the low temperature chamber.

4.3.2.4 High altitude.- The sextant and mount shall be subjected to a pressure of 3.44 ±0.08 inches of mercury absolute (corresponding to an altitude of 50,000 feet above sea level) for a continuous period of 4 hours. While at this pressure the instruments shall be operated and subjected to the Individual and Sampling tests specified herein. There shall be no failure as a result of this test. At the discretion of the testing activity, the Individual tests may be performed immediately after removing the sextant and mount from the pressure chamber.

4.3.2.5 High altitude - low temperature.- The sextant and mount shall be subjected to a combination of high altitude, as described above, and low temperature as described in the Low temperature exposure test, for a continuous period of 4 hours. While at this combined temperature and pressure, the instruments shall be subjected to the Individual and Sampling tests specified herein. There shall be no failure as a result of this test. This test may be made an additional part of the Low temperature exposure test. At the discretion of the testing activity, the Individual tests may be performed immediately after removing the sextant and mount from the pressure chamber.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T05:47Z
Check the source to verify that this is the current version before use.

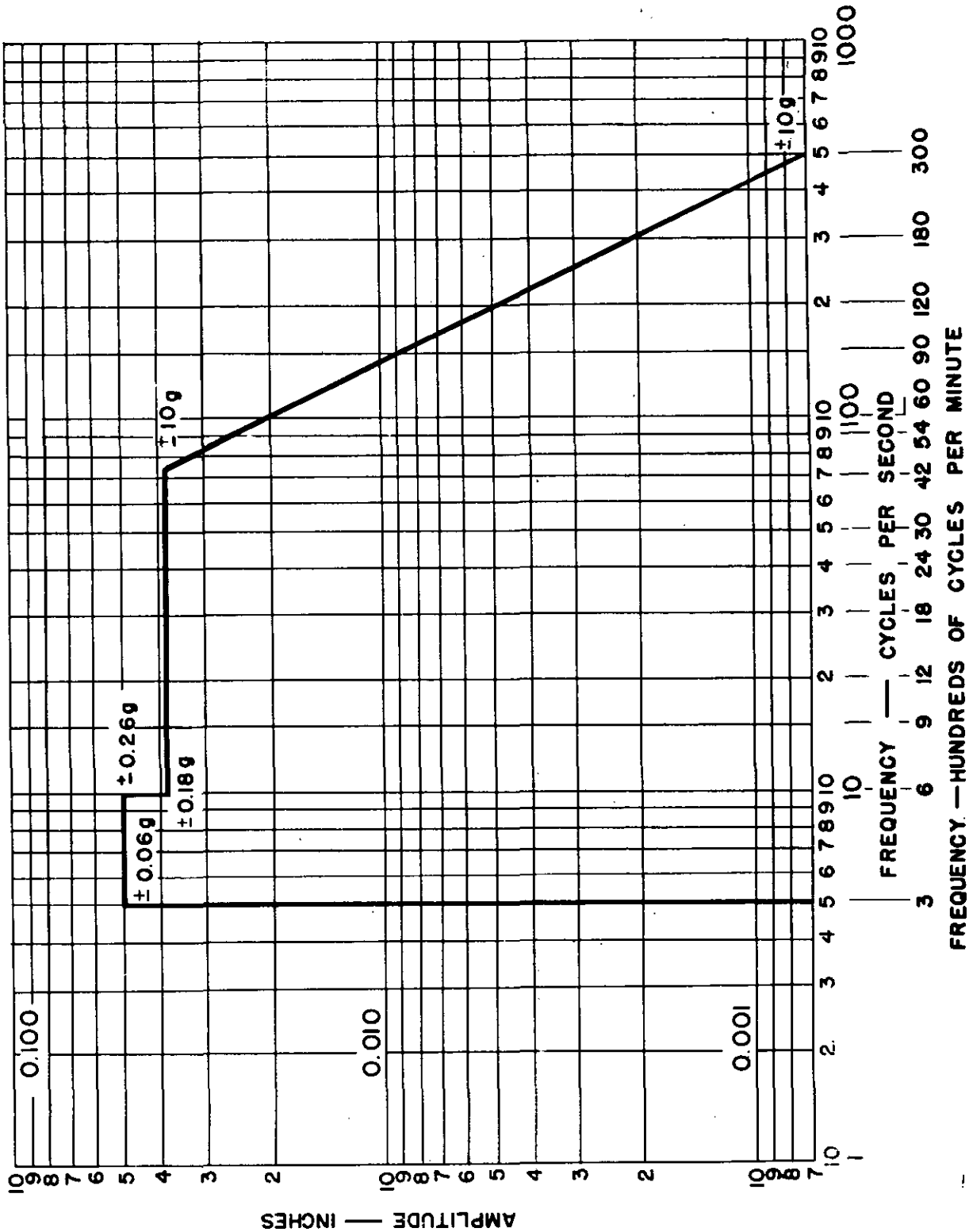


FIGURE 1. Range curve for Vibration tests

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T05:47Z
Check the source to verify that this is the current version before use.

4.3.2.6 Rain.- The sextant shall be subjected to a rain test in accordance with the Rain tests, Procedure I, of Specification MIL-E-5272. The sextant shall be mounted in the sextant mounting bracket which in turn is mounted in the plastic astro sighting window in the normal operating position inside the rain test chamber. The portion of the sextant which extends below the astro sighting window shall be protected from the rain. For a 30-minute period during the Rain test and with the sextant in the extended position, the sextant shall be protected from all rain except that which might enter through the mount. A flat plate may be fabricated for test purposes in lieu of the astro sighting window. The Rain test shall then be repeated with the mount cover closed. In mounting the sextant, external connections should be made to simulate installation conditions as near as possible. Duration of the test period shall be 8 hours, at the completion of which, the sextant shall be examined and no evidence of water penetration or damage shall be visible. At the end of the test with the sextant installed, the mount shall be examined, and there shall be no evidence of water damage to the sextant or mount. At the end of the test, with the mount cover closed, there shall be no evidence of water penetration or damage to the sextant or mount.

4.3.2.7 Humidity.- The Humidity test shall be performed in accordance with the Humidity tests, Procedure I, of Specification MIL-E-5272. The source of heat for the chamber shall be so arranged that the radiant heat shall not fall upon the specimen. The silica gel indicators shall not indicate the presence of moisture.

4.3.2.8 Fungus resistance.- The Fungus resistance test shall be in accordance with the Fungus resistance tests, Procedure I, of Specification MIL-E-5272.

4.3.2.9 Salt fog.- The Salt fog test shall be in accordance with the Salt spray test requirements of Specification MIL-E-5272. After removal of the salt encrustations, the sextant shall be subjected to the Individual tests specified herein.

4.3.2.10 Filter test.- Conformance to the paragraph in Section 3 entitled "Filters and shade glasses" is required.

4.3.2.11 Drop test.- With all components in place within the closed carrying case, no movement of the components shall be felt when the case is shaken vigorously. After one of the sample sextants and cases has been exposed to a temperature of 0°C for a period long enough to reach saturation, the sextant and case shall be dropped on a firm solid surface from a height of 30 inches in any attitude or position. The test shall be repeated with the other test sample after it has been saturated at a temperature of 71°C. No damage shall occur which would adversely affect subsequent operation. This test may be combined with the High and Low temperature tests if desired. As a result of this test the carrying case shall not be functionally damaged.

4.4 Acceptance tests.- The Acceptance tests shall consist of Individual tests and Sampling tests.

4.4.1 Individual tests.- Each sextant shall be subjected to the following tests, in the following order.

4.4.1.1 Examination of product.- Each sextant, mount, and case shall be inspected to determine conformance with this specification with respect to materials, workmanship, design requirements, performance, and envelope.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T05:47Z
Check the source to verify that this is the current version before use.

4.4.1.2 Scale error.- Each sample sextant shall be tested for scale error at five points; -1, 0, +1, 45, and 90 degrees of altitude with the bubble centered on the cross hairs, or at any other points selected at the discretion of the testing activity with both increasing and decreasing readings. The altitude error at any point shall not exceed 2 minutes of arc and the cross hairs shall not deviate laterally from the reference centers by more than 5 minutes of arc. The sextant and mount shall, in addition, be bench tested as a unit to determine true heading error caused solely by the instrument. True heading shall be checked each 90 degrees from zero to 360 degrees with azimuth set in to 1/10 degree, bubble in the center of the field as indicated by the cross hairs, and the cross hairs alined on the body or simulated body. Total error in true heading shall not exceed 1/4 degree.

4.4.1.3 Bubble cell curvature.- Each sample sextant shall be tested for corrections of bubble cell curvature. The error due to incorrect curvature of the bubble cell when using a bubble of approximately 5/64-inch diameter shall not exceed 2 minutes of arc at any point along the line of sight up to 2/3 of the distance from the center to the edge of the field.

4.4.1.4 Shade glass.- Each sample sextant shall be tested to determine the error caused by the shade glasses. The use of any shade glass shall not introduce secondary images in the line of sight or cause a change in the azimuth reading of the sextant of more than 3 minutes of arc in a horizontal plane or altitude reading of 30 seconds in a vertical plane. This test may be combined with the Scale error test.

4.4.1.5 Fixed setting.- The averager shall be operated for any desired period of time between 1 and 2 minutes on any altitude setting with the knob held stationary. Movement of the primary indicating element shall be not more than 0.010 inch. This test shall be performed three times using different operating periods and drum settings, arbitrarily chosen.

4.4.1.6 Slippage.- The averager shall be operated for the maximum possible time interval and the altitude setting noted after bringing the indices into coincidence. Without rewinding the mechanism, the knob shall be rapidly oscillated 5 degrees from its setting. After a minimum of five cycles, the indicator shall be brought into coincidence. The altitude setting then obtained from the averager shall not differ from that previously obtained by more than 2 minutes of arc. To eliminate the effect of backlash from this test, both altitude readings shall be obtained as the primary element of the averaging indicator is brought into alinement by rotation of the knob in the same direction as that of the start of the cycles. This test may be combined with other tests if desired.

4.4.1.7 Continuously changing settings (assembled sextant).- The averager shall be operated for any desired period of time during which the altitude settings change continuously and at a constant rate, through a range of not more than 2-1/2 degrees. The average altitude computed by the averager shall agree with the average of the initial and final altitude settings to within the following:

At the end of 30 seconds, ±5 minutes of arc;
At the end of 1 minute, ±3 minutes of arc; and
At the end of 2 minutes, ±2 minutes of arc.

This test shall be performed five times using different operating periods and altitude settings arbitrarily chosen. At least one test shall include initial and final drum settings differing by approximately 2-1/2 degrees. The continuously changing setting tests shall be performed on the averager prior to mounting on the sextant.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T05:47Z
Check the source to verify that this is the current version before use.

4.4.1.8 Variable time averager (VTA)-zero adjustment.- After the instrument has been rewound, and before the averaging mechanism is put into operation, oscillation of the knob through 10 degrees range of reading shall not move the element of the averaging indicator more than 0.010 inch for a series of five or more cycles.

4.4.1.9 Variable time averager (VTA)-zero setting.- After the averaging device has operated for a complete cycle of operations, it shall be rewound. When rewound, the indicator shall return to within ±0.010 inch of its zero index mark.

4.4.1.10 Variable time averager (VTA)-backlash in averaging mechanism.- After the instrument has run for the maximum time interval and before it is rewound, backlash in the averaging mechanism shall be determined by bringing the indicating elements and the index marker into coincidence first from one direction then from the opposite direction and noting the resulting sextant altitude readings. These readings shall not differ by more than 1 minute. To eliminate the effect of slippage all operations of this test shall be done slowly.

4.4.1.11 Maximum operating period.- The averager shall be operated three times for the maximum time interval. The length of time of each such period shall be 2 minutes ±4 seconds. This test may be combined with other tests if desired.

4.4.1.12 Sextant case.- The sextant carrying case shall be inspected for signs of any form of deterioration or loose fitting components which would make it unfit for normal use.

4.4.1.13 Dielectric strength.- With the lamps removed, the sextant and mount shall be subjected to a high potential test. When 500 volts rms, 60 cycle a-c power is applied for 1 minute between "A" and "B" of the receptacles, there shall be no breakdown of insulation.

4.4.1.14 Mount performance.- The mount crank shall be rotated for at least 1 minute in each direction at speed not to be in excess of 20 degrees of azimuth counter indication per second. Care shall be taken to limit starting accelerations to the maximum obtainable by hand operation. No slippage of the counter shall be permitted, and conformance to paragraph 3.3.2.2 is required. After the rotation has been accomplished, the difference between the counter and the azimuth scale against the lubbers line shall not exceed 0.10 degree.

4.4.2 Sampling tests.- The Sampling tests listed below shall be performed on 100 percent of the sextants until 5 consecutive good sextants are found. Then 1 sextant shall be selected at random from each of the next two sets of 50 sextants produced. Then 1 in 100 shall be selected at random, and the test procedure shall continue on this basis. However, as soon as a defective sextant is found, the following procedure shall be followed: A minimum set of 6 sextants about the defective sextant shall be selected and tested for the defect or defects found. However, the Inspector may have as many additional sextants tested as he desires to establish that the trouble has been overcome. This procedure shall be continued until a set of 6 is free of defects or until the entire set of 50 or 100 within which the original defective sextant was found is used up. When the set of 6 sextants free of defects is found, the 1-in-50 sampling phase for all tests shall be resumed, but beginning the set of 50 with the first sextant following the last sextant tested in the production sequence. If a set of 6 sextants free of defects is not found, the original 100 percent testing plan using all the tests shall be resumed. All sextants and mounts of the same type presented at one time shall be considered a lot for purposes of acceptance inspection and tests. For purposes of these tests, the term "sextant" shall be construed to include the sextant and carrying case.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T05:47Z
Check the source to verify that this is the current version before use.

4.4.2.1 Vibration error test.- The tests shall be conducted with the sextant installed in the mount in the "Up" position. The sextant shall be vibrated with a circular motion of 0.018- to 0.020-inch diameter in a plane inclined 45 degrees to the horizontal plane, and the frequency of vibration shall be varied uniformly from 500 to 3,000 cpm and returned once each hour for a 3-hour period. Creepage of the azimuth scale, as indicated by the counter, shall not exceed 0.25 degree during any 2-minute period of vibration.

4.4.2.2 Sextant and bubble cell temperature.- The sextant with the bubble adjusted in size to approximately 5/64 inch diameter, and bubble adjustment knob, in the maximum position, shall be subjected to a temperature of -54°C for a period of 4 hours. The operating controls and moving parts of the sextant shall function properly while at this temperature. The sextant shall then be subjected to a temperature of +70°C for a period of 4 hours without changing the setting of the bubble adjustment knob. No damage or leakage shall occur to the bubble cell as a result of this test. The operating controls and moving parts of the sextant shall function while at this temperature. The bubble size shall not change during the -54°C to +70°C run of this test more than 1/32 inch in diameter during any 50° change of temperature. At temperatures approximately -40°C and +40°C, the ease of formation and adjustment of the bubble and its damping, shall be substantially the same as at room temperature.

4.4.2.3 Nonfogging.- The sextant shall be placed in a test chamber which is capable of being sealed, and in which the temperature and relative humidity have been raised to 22°C and 95 - 100 percent, respectively. After 3 hours, the sextant shall be removed from the test chamber and placed in another chamber in which the temperature is -40°C. The pressure shall then be reduced to 9 inches Hg (approximately 30,000 feet altitude). After 1 hour in the cold chamber, the sextant shall be removed and so placed between heating elements that the metal case of the periscope tube receives the direct heat radiation. As the temperature of the sextant is raised, no fogging shall occur on the optical or glass surfaces, and the silica gel indicator shall not indicate the presence of moisture.

4.4.2.4 Illumination.- The sextant shall be tested to determine compliance with the requirements specified for illumination of bubble image and filters and shade glasses in Section 3.

4.4.2.5 Counter.- The sextant shall be tested to determine compliance with the requirements specified for the altitude counter and altitude range in Section 3.

4.4.2.6 Continuously changing settings (assembled sextant).- This test shall consist of the same test as described under Individual tests except that the tests shall be performed while the averager is assembled to the sextant.

4.4.2.7 Mount air flow leakage.- Conformance to paragraph 3.3.2.1 is required. This test shall be performed on the assumption that standard atmospheric pressure or higher would exist in the aircraft cabin and the outside air pressure would be 10 psi or less.

4.4.3 Rejection and retest.-

4.4.3.1 Deterioration.- Corrosion or deterioration of any internal or external components of the sextant, mount, or case which could in any manner prevent the equipment from meeting operational requirements during service life shall be cause for rejection.

4.4.3.2 When Sampling tests are specified on a number of items that are selected from a production run and one or more of this number fails to meet the specified test(s), acceptance of all items still on hand and subsequently produced will be withheld until the extent and cause of failure is determined. For operational reasons, Individual tests may be continued pending investigation of a Sampling test failure. However, final acceptance of the items on hand and subsequently produced is contingent upon the Inspector's decision regarding the over-all conformance of the product to specification

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T05:47Z
Check the source to verify that this is the current version before use.

requirements. When corrective action has been accomplished, all necessary tests shall be repeated. If investigation indicates that the defects may exist in items previously accepted, full particulars concerning the defects, including recommendations for correction, will be furnished to the contracting officer.

4.5 All parts, specimens, or assemblies destroyed in making tests required by this specification and drawings, to determine compliance with the specification and drawings, shall be in addition to the quantity specified in contract or purchase order and shall be furnished without increasing the cost of the contract or order.

5. PREPARATION FOR DELIVERY

5.1 Application.- The requirements of Section 5 apply only to direct purchases by or direct shipments to the Government.

5.2 Packaging and packing.-

5.2.1 Sextant.- The sextant shall be packaged in the carrying case and packed for shipment in accordance with Specification MIL-P-6064, type I, group X. The dehydrating agent shall be securely placed inside the sextant carrying case.

5.2.2 Mount.- The mount shall be packaged and packed for shipment in a type I container in accordance with group X of Specification MIL-P-6064.

5.3 Marking of shipments.- Interior packages and exterior shipping containers shall be marked in accordance with Standard MIL-STD-129, and shall also contain fragile markings. The nomenclature shall include: Federal Stock No. * (if no FSN available, leave space therefor).

5.3.1 Sextant.- The nomenclature for the sextant shall be as follows: Sextant, Aircraft, Periscopic, Variable Time Integrator, 28 Volts ac or dc, MS28011-1, (*Manufacturer's Part No.) (*Date of Manufacture), Specification MIL-S-5807C.

5.3.2 Mount.- The nomenclature for the mount shall be as follows: Mount, Periscopic Sextant, for MS28011 Sextant, (Manufacturer's Part No.), (*Date of Manufacture), Specification MIL-S-5807C.

*Applicable data to be entered by the contractor.

5.3.3 Reinspection dates (for Air Force use only).- Interior packages and shipping containers containing identical items shall bear the legend "REINSPECTION REQUIRED IF IN STORAGE AFTER (date)." The date entered shall be determined by adding, to the date of manufacture, the number of months specified in ANA Bulletin No. 405.

6. NOTES

6.1 Intended use.- The sextant covered by this specification is intended for use in aircraft to measure the angular altitude of a celestial body by reference to a bubble artificial horizon and to measure true heading of aircraft.

6.2 Superseding data.- The Office of Standardization has authorized the revision to this specification, previously fully coordinated, as an ASG limited coordination specification in view of its primarily aeronautical application.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T05:47Z
Check the source to verify that this is the current version before use.

6.3 Ordering data.- When ordering the periscopic sextant and periscopic sextant mount, the procuring activity should order the two items separately, using a different stock number for each item.

6.3.1 Preproduction tests.- It is expected that the contract or purchase order will specify that two sextants, mounts and case as applicable, will be required as preproduction samples and that these preproduction samples will be subjected to the Preproduction tests to determine compliance with the requirements of this specification. The invitation for bids and the contract should specify the point of inspection for these tests. For Air Force contracts, Preproduction test reports should be in accordance with Specification MIL-T-9107.

6.3.1.1 It is to be understood that items supplied under contract will be identical in every respect to the sample tested and found satisfactory, except for changes previously approved by the Government. Any unapproved changes from the samples will constitute cause for rejection.

6.4 Definitions.-

6.4.1 Amplitude.- Whenever the word "amplitude" is specified, it will mean the extent of motion as measured from one extreme to the opposite extreme.

PATENT NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Custodians:
Navy - Bureau of Aeronautics
Air Force

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T05:47Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

DD, FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863 WHICH IS OBSOLETE

PLATE NO. 15419

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T05:47Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE NAVY

POSTAGE AND FEES PAID
NAVY DEPARTMENT

OFFICIAL BUSINESS

CHIEF, BUREAU OF NAVAL WEAPONS
ENGINEERING DIVISION
ATTN: CODE RREN-5
DEPARTMENT OF THE NAVY
WASHINGTON, D. C. 20360

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T05:47Z
Check the source to verify that this is the current version before use.